Lin Wang

PhD Candidate, Department of Electrical & Computer Engineering, Texas A&M University

Wireless Networks · ML Systems · Stochastic Optimization & Scheduling · Performance Engineering

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EDUCATION

Texas A&M University — Ph.D. in Electrical Engineering (Advisor: I-Hong Hou) *Aug 2024 – May 2028 (expected)*

Coursework: Stochastic Processes, Convex Optimization, Machine Learning, Deep Learning, Reinforcement Learning, Computer Networks

Nankai University — M.Sc. in Electrical Engineering (Communications) *Sep 2021 – Jun 2024*

GPA 4.0/4.0

Nankai University — B.Sc. in Electrical Engineering (Communications) *Sep 2017 – Jun 2021*

Coursework: Communication Theory, Signals & Systems, Signal Detection & Estimation, Cellular/Mobile Communications (OFDM, MIMO), Satellite Communications

RESEARCH INTERESTS

Wireless and radio access networks; real-time scheduling and resource allocation; stochastic modeling, optimization, and control for large-scale networked systems; machine-learning systems and LLM inference; reinforcement learning / ML for network and systems optimization.

RESEARCH EXPERIENCE

Graduate Research Assistant — Texas A&M University *Aug 2024 – Present*

Stochastic Modeling & Scheduling for Wireless Networks (NSF project) *Aug 2024 – Present*

- Formulated resource allocation in unreliable multichannel wireless networks as a **Markov Decision Process** under a second-order (mean–variance) model; derived online decision policies with provable stability guarantees using stochastic processes and queueing theory.
- Proved the proposed policy achieves the theoretical optimal bound; validated in simulation that performance closely tracks the bound while outperforming strong baselines (Whittle Index, Max-Weight).
- Built reproducible Python simulation and benchmarking pipelines, running large-scale experiments on Slurm-managed HPC clusters. **Two first-author ACM MobiHoc papers.**

AI Infrastructure — Adaptive Control for LLM Serving *Jan 2026 – Present*

- Integrated a dynamic speculative-decoding controller into the vLLM serving engine, adapting speculation strategy in real time to time-varying request arrivals in an online serving system.
- Formulated adaptive control of speculation length as a constrained sequential decision problem; trained a reinforcement-learning policy that responds to fluctuating system dynamics to minimize end-to-end latency.
- Implemented and evaluated a range of speculative-decoding methods (draft-model, EAGLE-3, DFlash); profiled serving with Linux perf, FlameGraph, Nsight Systems/Compute, and PyTorch Profiler to attribute latency to kernel execution, batching, and KV-cache effects.

Real-Time Scheduling for 5G Radio Access Networks (O-RAN) *Aug 2025 – Present*

- Developed and deployed sub-millisecond radio scheduling and control modules on the EdgeRIC real-time RIC platform, implementing RT-E2 interfaces and μApp pipelines for AI-driven RAN optimization.
- Designed low-latency policy-execution engines and cross-layer feedback loops that improve network-control responsiveness while sustaining high throughput under tight timing constraints.

INDUSTRY EXPERIENCE

Technical Student Intern — NXP Semiconductors *Jul 2023 – Jan 2024*

- Deployed quantized computer-vision models (image classification, 3D object detection) on i.MX93 NPU edge devices using TensorFlow Lite with hardware delegate acceleration, under strict latency and memory budgets.

- Profiled inference latency and bottlenecks (warm-up vs. steady-state) and optimized detection-and-tracking pipelines; integrated them into production embedded software with a cross-functional team.

TEACHING

Teaching Assistant — Texas A&M University *Fall 2026*

ECEN 424/500: Fundamentals of Networking.

PUBLICATIONS

- **Lin Wang** and I-Hong Hou. "Understanding the Fundamental Trade-Off Between Age of Information and Throughput in Unreliable Wireless Networks." **ACM MobiHoc 2025**.
- **Lin Wang** and I-Hong Hou. "TSDM: A Scheduling Policy for Joint Throughput–AoI Optimization in Multichannel Wireless Networks." **ACM MobiHoc 2026**.
- Ruili Nan, Guiling Sun, Bowen Zheng, and **Lin Wang**. "Adaptive Deep Learning Network for Image Reconstruction of Compressed Sensing." *Signal, Image and Video Processing*, 18(2):1463–1475, 2024.

PROFESSIONAL SERVICE

- **Reviewer:** IEEE/ACM Transactions on Networking (ToN); IEEE Transactions on Mobile Computing (TMC); IEEE Open Journal of the Communications Society (OJ-COMS).
- **Shadow PC:** ACM SIGMETRICS 2026.

HONORS & AWARDS

- Merit Graduate Fellowship, Texas A&M University (2024)
- National Scholarship, Nankai University (2024)
- National Third Prize, "Challenge Cup" National College Students Innovation & Entrepreneurship Competition
- National Second Prize, National Undergraduate Electronics Design Contest (2019)

SELECTED PROJECTS

- **Smart Agriculture IoT System** (Nankai University): built an NB-IoT wireless sensing system on STM32 with end-to-end sensor-to-cloud telemetry and multi-parameter field decision logic.
- **Wireless EV Charging System** (Electronics Design Contest): designed a dynamic wireless charging system on MSP432 with auto-tracking, timed startup, and the power/firmware stack.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Bash; Linux, Git, REST APIs, CRUD / HTTP services
- **Performance & Systems:** Linux perf, FlameGraph, Nsight Systems/Compute, PyTorch Profiler, NVTX; benchmarking, bottleneck analysis, Slurm & HPC/GPU clusters, real-time systems, edge/on-device deployment
- **ML & Modeling:** LLM inference & serving (vLLM, llama.cpp), deep learning, reinforcement learning, PyTorch, TensorFlow Lite; MDPs, convex & stochastic optimization, queueing theory
- **Wireless & Communications:** 5G/LTE, O-RAN & RIC (RT-E2, μApps), OFDM, MIMO, communication theory, radio resource scheduling, Age of Information (AoI)
- **Data Analysis:** pandas, NumPy, Matplotlib; reproducible experiment design, trace/log analysis, visualization & reporting